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PROJECT REPORT ON

CYBERSHIELD AI: INTELLIGENT CYBER INCIDENT DETECTION AND SAFETY MANAGEMENT PORTAL

(SIH Problem Statement ID: SIH25183)

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ABSTRACT

Cyber attacks against defence infrastructure are escalating globally, requiring sophisticated response mechanisms. The existing cyber incident reporting and log analysis mechanisms suffer from fragmented workflows, absence of intelligent threat analysis, and poor real-time communication. This project, developed in response to Smart India Hackathon 2025 Problem Statement SIH25183 (Ministry of Defence), presents CyberShield AI -- an intelligent, full-stack web portal that integrates Artificial Intelligence for automated cyber incident detection, log analysis, and safety management.

The system prototype is built on a modern React/Vite stack with a Supabase PostgreSQL backend. It uses Row-Level Security for strict data isolation, and Google Gemini AI models for natural language processing and threat analysis. Key features include a defence personnel incident reporting dashboard, an AI-powered risk scoring engine, an Automated Threat Intelligence (ATI) Log Analyzer, an AI chatbot (CyberBot) for immediate guidance, a network safety alert broadcasting system, and a secure administrative command centre.

Keywords: Cybersecurity, Artificial Intelligence, Incident Reporting, Phishing Detection, Gemini AI, React, Supabase, Smart India Hackathon.

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CHAPTER 1

INTRODUCTION

1.1 Background

The digital infrastructure of the Armed Forces and the Ministry of Defence is a critical asset. While modern digitisation has streamlined operations, it has simultaneously exposed defence networks to a growing landscape of sophisticated cyber threats, including Advanced Persistent Threats (APTs) and zero-day vulnerabilities. Military networks face millions of anomalous events daily, spanning from phishing attacks on personnel to coordinated intrusion attempts.

Recognising the criticality of defence cybersecurity, there is a need for robust, air-gapped tools to enable defence personnel to securely report cyber incidents. The current processes face significant challenges: reporting workflows can be cumbersome, there is no real-time intelligent analysis of raw server logs, and the communication loop during active threats is slow. The Smart India Hackathon (SIH) 2025 identified this gap through Problem Statement ID SIH25183, which calls for the development of an Intelligent Cyber Incident Detection and Safety Management Portal.

1.2 Problem Statement

SIH25183 -- AI-Enabled Cyber Incident & Safety Web Portal for Defence. The problem statement requires the development of a highly secure, intelligent web portal capable of: receiving cyber incident reports securely; automatically analysing raw security logs and reported incidents using AI to determine risk levels and categorise threats; providing real-time containment alerts; and enabling authorities to monitor and respond to incidents efficiently.

1.3 Objectives

- To design and develop a secure, zero-trust web portal that enables defence personnel to report cyber incidents and administrators to analyse threats.
- To integrate AI for automated incident risk scoring, intelligent categorisation, and raw log analysis (ATI).
- To implement a real-time safety alert broadcasting system that enables authorities to issue network

containment advisories.

- To build an administrative analytics dashboard with visual representations of incident trends, severity distributions, and risk metrics.
- To demonstrate a prototype deployable in an air-gapped on-premises military cloud.
- To implement strict Role-Based Access Control (RBAC) and Row-Level Security (RLS) to ensure data isolation.

1.4 Scope of the Project

The scope encompasses: Secure User Management (Zero-Trust, RBAC); End-to-end Incident Lifecycle Management (submission, AI analysis, admin triage, resolution); AI capabilities including the Automated Threat Intelligence (ATI) Log Analyzer and risk scoring; Network Safety Alert Broadcasting; and Analytics dashboards.

CHAPTER 2

LITERATURE SURVEY

2.1 Review of Existing Technologies

2.1.1 National Cyber Crime Reporting Portal (cybercrime.gov.in)

The Government of India's official portal allows citizens to file complaints related to cybercrime, particularly financial fraud and crimes against women and children. The portal supports online FIR filing and tracks complaints through a reference number. Limitations: The portal uses a multi-page, text-heavy form that is not user-friendly. There is no AI-assisted analysis, no chatbot for guidance, and no mechanism for real-time safety alerts.

2.1.2 CERT-In Incident Reporting

The Indian Computer Emergency Response Team (CERT-In) accepts incident reports via email and a web form. It publishes advisories and vulnerability notes. Limitations: The reporting process is designed for IT professionals, not ordinary citizens. There is no automated triage, no risk scoring, and no public-facing dashboard.

2.1.3 International Systems

IC3 (FBI, USA) accepts cybercrime complaints via an online form but provides no real-time analysis or AI features. Action Fraud (UK) provides an online reporting tool with some automated categorisation but no AI chatbot. ACSC (Australia) provides a ReportCyber portal with structured forms but no AI integration.

2.2 Drawbacks of the Existing System

S.No.	Drawback	Impact
1	No AI risk assessment	Incidents need manual review
2	No phishing tool	Citizens cannot verify links
3	No AI chatbot	No immediate advice
4	No real-time alerts	Cannot warn citizens fast
5	Poor UI/UX	Complex forms stop reporting
6	No analytics	Decision-makers lack insights
7	No DB security	Data isolation uses app logic

CHAPTER 3

PROPOSED SYSTEM

3.1 System Overview

The proposed system, CyberShield AI, is a full-stack web application designed with a zero-trust architecture:

Defence Personnel Portal: A secure dashboard where military personnel can report cyber incidents (like suspected malware or phishing) through a guided form, track report status, interact with CyberBot (AI assistant), and receive urgent network safety alerts.

Admin Command Centre (ATI): A restricted dashboard where administrators monitor all incidents, access visual analytics, broadcast safety alerts, and most importantly, use the Automated Threat Intelligence (ATI) Log Analyzer to parse raw firewall/server logs for zero-day anomalies.

The system prototype uses modern web technologies (React 18, TypeScript, Tailwind) backed by a relational database with RLS (Supabase) and Google Gemini AI for advanced analytics.

3.2 Advantages over Existing System

Feature	Existing System	CyberShield AI
Reporting Form	Complex multi-page	Streamlined military format
Log Analysis	Manual Regex/Grep	ATI Log Analyzer (AI)
Categorisation	Manual	AI-suggested APT/Malware type
Victim Guidance	Static documents	Context-aware AI chatbot
Safety Alerts	Emails/Notices	Real-time broadcast + audio
Access Control	Basic permissions	Zero-Trust DB-level RLS
Deployment Target	Public Cloud	Air-Gapped Military Network

CHAPTER 4

SYSTEM DESIGN

4.1 System Architecture

The application follows a modern cloud-native three-tier architecture using React and PostgreSQL (via Supabase) to deliver a highly responsive and secure user experience.

SYSTEM ARCHITECTURE

```
|-- Presentation: React 18 (Personnel & Admin Portals)
|-- Application: Supabase REST API, Auth
|-- Data: PostgreSQL + RLS
|-- AI Core: Google Gemini (Risk Analysis, Chatbot)
```

4.2 Module-wise Breakdown

The application is divided into 8 functional modules:

Module 1: Authentication & Zero-Trust

Role-based access, automatic profile mapping, and strict route protection.

Module 2: Personnel Incident Reporting

Guided form capturing military-specific incident types (APT, Insider Threat, Network Malware).

Module 3: AI Risk Scoring Engine

Automated triage and severity assessment (0-100 scale).

Module 4: Automated Threat Intelligence (ATI)

Admin module to paste raw firewall logs. AI parses logs and outputs containment protocols.

Module 5: AI Chatbot (CyberBot)

Floating widget providing immediate, context-aware cybersecurity guidance.

Module 6: Admin Command Centre

Dashboard with KPI cards, searchable incident table with detail modals, visual analytics.

Module 7: Safety Alert Broadcasting

Admin CRUD for network alerts. Personnel page with synthesised audio notifications.

Module 8: Phishing/URL Detector

URL/email analysis returning verdict, threat score, and red flags.

4.3 Database Design

Table	Key Columns	Purpose
profiles	id, email, full_name, role	User profile data
incidents	id, title, type, severity	Cyber incident reports
alerts	id, title, content, severity	Public safety alerts
ai_analyses	id, risk_score, keywords	AI analysis results

4.4 Technology Stack

Layer	Technology	Purpose
Frontend	React 18.3 + TS 5.5	Component-based UI
Build Tool	Vite 5.4	Fast dev server
CSS	Tailwind CSS 3.4	Utility-first styling
Routing	React Router 7.17	Client navigation
Charts	Recharts 3.8	Data visualisation
AI Engine	Gemini (2.5-flash)	NLP, risk analysis
Backend	Supabase (Postgres)	DB, Auth, RLS
Deployment	Vercel	CDN hosting

CHAPTER 5

IMPLEMENTATION

5.1 Development Setup

The project was initialised using the Vite build tool with the React-TypeScript template:

```
npx create-vite@latest cybersafe-portal --template react-ts
npm install @supabase/supabase-js @google/generative-ai
npm install -D tailwindcss
```

5.2 Authentication & Zero-Trust (RBAC / RLS)

Implemented using a Context Provider pattern. `ProtectedRoute.tsx` enforces strict role-based access, isolating defence personnel views from admin capabilities. At the database level, Supabase Row-Level Security (RLS) enforces strict data isolation. For example, the Global Threat Map dynamically filters data: an ordinary citizen can only see their own reported threats on the map, whereas an Admin account bypasses this restriction to visualise all active threats across the network simultaneously. This zero-trust architecture ensures that even if an attacker gains citizen-level access, they cannot harvest other incident reports.

5.3 Incident Reporting

The interactive form collects structured data with fields for title, military category (e.g., APT), severity level, description, platform, and date. The AI Risk Scoring Engine processes data to assign priority before alerting admins.

5.4 AI Risk Scoring Engine

A system that analyses the incident narrative to assign a risk score from 0-100, providing immediate triage classification (Low, Medium, High, Critical) for administrators.

5.5 Automated Threat Intelligence (ATI) Log Analyzer

A specialised admin tool designed to ingest raw, unstructured security logs (firewall, server, IDS). The AI engine parses the logs, identifies anomalies or attack patterns, and automatically suggests military-grade containment protocols (e.g., block IP, isolate VLAN).

5.6 AI Chatbot (CyberBot)

A floating widget communicating with the AI backend using multi-turn conversation. Features include context-aware cybersecurity guidance tailored for defence personnel.

5.6 Admin Command Centre

Dashboard with real-time KPI cards. AdminIncidents.tsx provides a searchable table with clickable rows opening detail modals showing full AI analysis. AdminAnalytics.tsx provides charts for trends, severity, and categories.

5.7 Safety Alerts

Administrators create alerts. Citizens view them with a synthesised notification sound using the Web Audio API with an exponential frequency sweep.

5.8 Challenges & Solutions

Challenge	Solution
Silent database failures	Explicit error checking added
Gemini API rate limiting	Model fallback chain + retry delay
DB schema mismatches	ALTER TABLE statements added
Hardcoded risk scores	Supabase RPC function for dynamic data

CHAPTER 6

RESULTS AND DISCUSSION

6.1 System Outputs

The CyberShield AI prototype was successfully completed. Key outputs include:

- Landing Page: Cyberpunk-themed interface tailored for defence personnel.
- Personnel Dashboard: Role-appropriate navigation with quick-access incident reporting.
- ATI Log Analyzer: Successfully parses raw logs and outputs structured threat mitigation steps.
- CyberBot Chatbot: Floating widget responding to queries with cybersecurity context.
- Phishing Detector: Colour-coded verdict cards with detailed red flags.
- Admin Analytics: Dashboard with visual metrics and full incident audit trails.

6.2 Performance Metrics

Metric	Observed Value
Page Load	< 2 seconds (Vercel)
Supabase Query	50-150 ms
Gemini API Time	1-3 seconds
Lighthouse Score	85+ Performance, 90+ Accessibility
RLS Enforcement	100% strict data isolation

CHAPTER 7

CONCLUSION AND FUTURE WORK

7.1 Conclusion

The CyberShield AI portal was successfully designed and deployed for Smart India Hackathon 2025. It demonstrates that integrating Artificial Intelligence into cyber incident management improves efficiency and accessibility.

- Production-deployed web portal built with React, Supabase, and Tailwind.
- Three AI-powered features via Google Gemini (scoring, phishing, chatbot).
- Secure, role-based system with database-level RLS.

7.2 Future Enhancements

- Mobile Application: Native Android/iOS apps for broader accessibility.
- Integration with CCTNS: Direct API integration for incident escalation.
- Multi-language Support: Localisation into regional Indian languages.
- File Upload: Cloud-based evidence attachment using Supabase Storage.
- Push Notifications: Browser push and SMS alerts using Twilio.

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END OF REPORT

*CyberShield AI: Intelligent Cyber Incident Detection
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